

Algorithmic Generation and Exploitation of Topology-Aware Models for Semiexperimental Structure Refinement

Vincenzo Barone^{a)}

INSTM, National Interuniversity Consortium for Materials Science and Technology, via G. Giusti 9, 50121 Firenze, Italy

Semiexperimental equilibrium structures combine high-resolution rotational spectroscopy with computed vibrational and electronic corrections. The least-squares solution of a chosen refinement model is now routine; the less formalized step is the generation of that model. Active coordinates, symmetry, weakly determined directions, constraints, and parameter classes are still often selected by expert intervention, especially for cyclic, fused, and bridged molecular graphs.

This paper introduces MORPHEUS (Model Optimization and Refinement from Perceived Hierarchy, Equivalence, Uncertainty, and Symmetry), a framework that turns model formulation into an auditable topology-perception problem. The resulting models are not merely topology-aware: the active representation and all constraint or parameter-class relations are topology-derived, symmetry-consistent, rank-consistent, and recorded. Molecular topology defines the primitive coordinate space, symmetry selects the admissible active subspace, isotopic coverage determines which directions are supported, and electronic-structure reference information enters only through explicit constraints, predicates, or parameter classes.

Two equivalent active spaces are available: symmetry-adapted generalized internal coordinates built from homogeneous primitive families, and symmetry-adapted Cartesian displacements after removal of translations and rotations. The chemical interface remains internal-coordinate based: constraints, predicates, and parameter classes are expressed as primitive bond, angle, dihedral, out-of-plane, ring, or local-frame relations and then projected onto the selected active basis. Validation on acyclic, cyclic, planar aromatic, fused, bridged, and flexible benchmarks shows that the same rank, effective dimension, and rotational residuals are obtained without manual coordinate-model formulation, while final structural parameters and uncertainties remain chemically interpretable. A 49-atom testosterone scale check further shows how quantum chemical equilibrium geometries, accelerated vibrational corrections, and MORPHEUS class-constrained model generation can be combined for large molecular systems.

Keywords: semiexperimental equilibrium structures, rotational spectroscopy, generalized internal coordinates, least squares, Kraitchman equations

I. INTRODUCTION

Accurate equilibrium molecular structures are central to rotational spectroscopy, thermochemistry, force-field development, and the validation of electronic-structure methods.^{1,2} Despite decades of progress in semiexperimental refinement, model construction remains largely a manual activity. Structure determination from experimental observables is also a coordinate-dependent inverse problem: the data constrain a Cartesian geometry, but the fitted variables determine conditioning, constraints, correlations, and uncertainties. Rotational spectroscopy is a stringent example because rotational constants probe the molecular mass distribution with high precision, whereas semiexperimental (r_e^{SE}) structures require vibrational and sometimes electronic corrections to convert ground-state constants into equilibrium information.³⁻⁶

Modern fitting programs and workflows⁷⁻⁹ have established robust statistical principles: fit well-conditioned observables, include theoretical predicates when experimental information is incomplete, and propagate co-

variance to derived structural parameters.¹⁰ Their common prerequisite is that the refinement model has already been formulated. That prerequisite is not innocent: the active coordinates, symmetry restrictions, isotopic-coverage assumptions, reference information, constraints, and parameter classes define the statistical problem being solved.

This issue is modest for small acyclic molecules, but becomes central for rings, fused systems, and bridged frameworks. A molecular graph with cycles is not a tree; different reference paths can make one distance, angle, or torsion primary and an equivalent one derived. For such systems, construction of a symmetry-consistent non-redundant refinement model is itself a major part of the semiexperimental analysis. The target of the present work is therefore not a new least-squares observable or a molecule-specific improvement in residuals, but an algorithmic route from perceived topology to a reproducible statistical refinement model.

Recent automated workflows have extended semiexperimental refinement by combining computed starting geometries, quantum chemical (QC) vibrational corrections, and reduced-dimensionality models.¹¹ MORPHEUS (Model Optimization and Refinement from Perceived Hierarchy, Equivalence, Uncertainty, and Symmetry) uses such data as upstream inputs while making the re-

^{a)} Electronic mail: vincebarone52@gmail.com

finement model itself topology-derived. Reference geometries from QC computations or from fragment libraries (e.g., LCB25¹²) enter only as chemically explicit constraints, predicates, or parameter classes; the fitted variables remain the algorithmically generated non-redundant active coordinates. This separation is important for larger molecules. Accurate QC equilibrium geometries and vibrational corrections are prerequisites for a meaningful comparison with ground-state rotational constants; the framework adds the missing step of converting those ingredients into an explicit, rank-audited refinement model without molecule-specific coordinate design.

The resulting topology-derived workflow starts from Cartesian coordinates. A first set of primitive generalized internal coordinates (GICs) is built, redundancies within homogeneous coordinate families are removed, symmetry is assigned, and the totally symmetric active space is defined. The advisor then analyzes isotope coverage, symmetry orbits, chemical descriptors, and reference data to generate compatible constraints, predicates, and parameter classes. Two active spaces are available: symmetry-adapted GICs and symmetry-adapted Cartesians obtained after removing translations and rotations. Constraints remain primitive internal-coordinate relations and are projected onto either active basis by the Wilson matrix or the corresponding derivative operator.¹³

The rotational semiexperimental problem is used here as a demanding test because it exposes both rank deficiencies and chemically meaningful constraints. Glycolaldehyde is retained as a standard acyclic case. Cyclopentadiene, nitrobenzene, azulene, and norcamphor test ring closures, planar-component selection, fused topology, bridged topology, and hydrogen-frame constraints. The maleic, phthalic, and succinic anhydrides form a homologous literature-data series with different degrees of isotopic completeness and different auxiliary relations. Two glycine conformers test reduced-dimensionality constraints, and testosterone closes the study as a large-molecule scale demonstration.

II. THEORY AND METHOD

A. Overview of the MORPHEUS Framework

The overall architecture is summarized in Figure 1. The key separation is between the chemical model, the numerical active basis, and the least-squares solution. Chemical information is always expressed in primitive internal coordinates; the active variables may be GIC amplitudes or symmetry-adapted Cartesian displacements. This separation is what makes the generated model auditable: the same primitive relation has the same chemical meaning regardless of the basis used to solve the refinement.

B. Primitive and Work Coordinates

The coordinate construction starts from Cartesian geometry and perceived topology. Bonded pairs and ring connectivity define a primitive set of internal coordinates: stretches, valence angles, torsions, out-of-plane deformations, linear bends, and ring-specific primitives. Redundancy is intentional at this stage because it lets topology, not a manually chosen reference path, decide which combinations survive. For cyclic systems, endocyclic torsions and analogous endocyclic valence-angle combinations provide ring coordinates within the GIC formalism, so monocyclic, fused-ring, bridged, and polycyclic systems are handled by the same machinery.¹⁴ This primitive layer is also the chemical interface for constraints, predicates, and parameter classes.

C. Coordinates for Parameter Refinement

The active space must be non-redundant, symmetry-consistent, and compatible with chemically meaningful constraints. In the GIC route, topological primitives are reduced within homogeneous families, so stretches, bends, torsions, out-of-plane motions, and ring coordinates cannot replace one another during rank reduction.^{15–18} In the Cartesian route, translations and rotations are removed from the Cartesian displacement space. In both cases symmetry selects the totally symmetric active block and completes primitive constraints over equivalent orbits.

At a reference geometry $\mathbf{x}_0$, primitive displacements and Cartesian displacements are related to first order by the Wilson matrix¹³

$$\Delta \mathbf{s} = \mathbf{B}_s \Delta \mathbf{x}, \quad \mathbf{B}_s = \left. \frac{\partial \mathbf{s}}{\partial \mathbf{x}} \right|_{\mathbf{x}_0}. \quad (1)$$

For each homogeneous primitive family α , the Wilson block

$$\mathbf{B}_{s_\alpha} = \frac{\partial \mathbf{s}_\alpha}{\partial \mathbf{x}}. \quad (2)$$

is submitted to a singular-value decomposition,¹⁹

$$\mathbf{B}_{s_\alpha} = \mathbf{U}_\alpha \boldsymbol{\Sigma}_\alpha \mathbf{V}_\alpha^T, \quad (3)$$

The retained left-singular vectors define the row-orthonormal transformation

$$\mathbf{R}_\alpha = \mathbf{U}_{\alpha,r}^T, \quad (4)$$

where the subscript r denotes the retained singular subspace. The non-redundant coordinates for family α are then

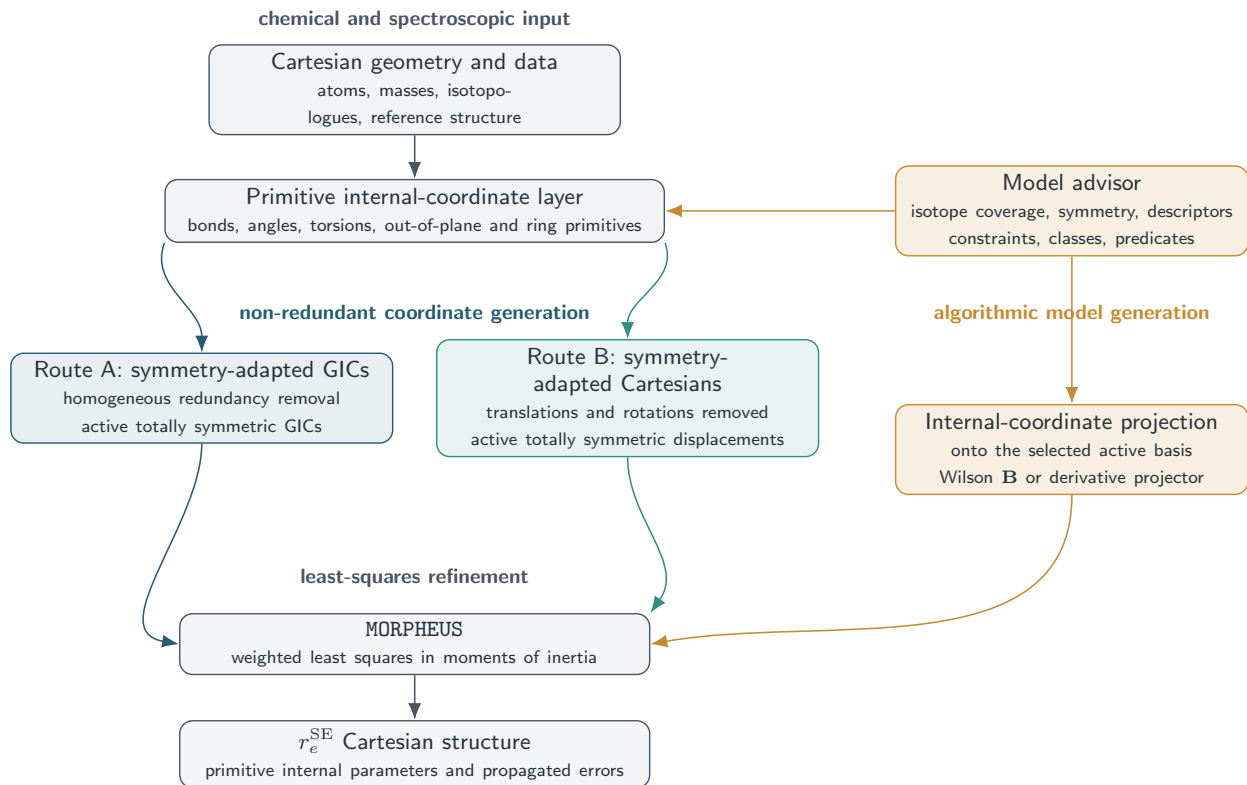


FIG. 1. Architecture of the MORPHEUS model-generation framework.

$$\Delta \mathbf{q}_\alpha = \mathbf{R}_\alpha \Delta \mathbf{s}_\alpha, \quad \mathbf{B}_{q_\alpha} = \mathbf{R}_\alpha \mathbf{B}_{s_\alpha}. \quad (5)$$

The retained rank in each family gives the number of coordinates in that family. Stacking the blocks gives

$$\Delta \mathbf{q} = \mathbf{R} \Delta \mathbf{s}, \quad \mathbf{B}_q = \mathbf{R} \mathbf{B}_s. \quad (6)$$

The block-diagonal structure preserves chemical homogeneity. A stretch can therefore not remove a torsion from the model, and a ring coordinate cannot be lost because a different primitive family happened to enter the rank test first. After family-wise reduction, residual dependencies are pruned with deterministic topological tie breaking and the final basis is checked against the expected vibrational rank. Cartesian displacements associated with GIC corrections use a rank-controlled minimum-norm pseudoinverse with explicit singular-value thresholds. The symmetry-Cartesian route uses the same selection, constraint projection, and covariance propagation.

D. Constraints, Predicates, and Parameter Classes

Structural relations are specified in primitive internal coordinates,

$$\mathbf{s} = (s_1, s_2, \dots, s_m)^T, \quad (7)$$

and may be supplied by the user or generated by the advisor. The same projector is used for constraints, predicates, classes, and error propagation, so changing from GICs to symmetry-adapted Cartesians changes the numerical basis, not the primitive chemical relations that define the model. Primitive-coordinate predicates are parsed canonically, and reversible angle or torsion orderings are normalized before projection.

a. Hard constraints. Hard constraints are exact equations imposed throughout the refinement:

$$s_i = s_i^0, \quad (8)$$

equalities,

$$s_i - s_j = 0, \quad (9)$$

or general linearized functions,

$$\mathbf{c}(\mathbf{s}) = \mathbf{d}. \quad (10)$$

At a trial geometry the corresponding first-order condition is

$$\mathbf{C}_s \Delta \mathbf{s} = \mathbf{b}_c, \quad \mathbf{C}_s = \left. \frac{\partial \mathbf{c}}{\partial \mathbf{s}} \right|_{\mathbf{s}_0}, \quad \mathbf{b}_c = \mathbf{d} - \mathbf{c}(\mathbf{s}_0). \quad (11)$$

Projection to the active space gives

$$\mathbf{C}_A \Delta \mathbf{a} = \mathbf{b}_c, \quad \mathbf{C}_A = \mathbf{C}_s \mathbf{B}_s \mathbf{T}_A. \quad (12)$$

The solver keeps all compatible motions by writing the allowed correction as

$$\Delta \mathbf{a} = \Delta \mathbf{a}_0 + \mathbf{N} \Delta \mathbf{p}, \quad \mathbf{C}_A \mathbf{N} = \mathbf{0}, \quad (13)$$

where $\Delta \mathbf{a}_0$ is a minimum-norm particular solution and $\mathbf{N}$ spans the null space. The least-squares problem is solved only for $\Delta \mathbf{p}$.

Symmetry-equivalent primitives are constrained as complete orbits before the null-space projector is formed. Reduced-dimensionality models use the same projection, allowing unsupported hydrogen frames to be frozen or tied without altering the covalent topology used for GIC generation.

b. Predicate observations. Predicates are soft structural information: they add weighted residuals rather than removing degrees of freedom.¹⁰ A primitive predicate has the form

$$f(\mathbf{s}) = f^{\text{ref}} \pm \sigma_f, \quad (14)$$

and contributes

$$r_f = \frac{f(\mathbf{s}) - f^{\text{ref}}}{\sigma_f} \quad (15)$$

to the residual vector. Its Jacobian with respect to the independent least-squares variables is

$$\mathbf{J}_f = \frac{1}{\sigma_f} \frac{\partial f}{\partial \mathbf{s}} \mathbf{B}_s \mathbf{T}_A \mathbf{N}. \quad (16)$$

c. Parameter classes. Parameter classes make several primitive quantities share one correction. If class c contains primitives $i \in c$, the allowed correction is

$$\Delta s_i = \Delta p_c, \quad i \in c. \quad (17)$$

Class definitions are converted into linear relations or Jacobian compression. They apply to corrections rather than final values, so related quantities may retain different reference values while sharing a fitted offset. After convergence, the covariance is expanded back to primitive and topological parameters.

E. Semiexperimental Observables

For each isotopologue s , the measured ground-state rotational constants are converted into equilibrium rotational constants by removing vibrational and, when necessary, electronic contributions,^{3–6}

$$B_{\alpha,e}^{(s)} = B_{\alpha,0}^{(s)} - \Delta B_{\alpha,\text{vib}}^{(s)} - \Delta B_{\alpha,\text{elec}}^{(s)}, \quad \alpha \in \{a, b, c\}. \quad (18)$$

The correction terms may come from any electronic-structure or composite protocol and are supplied with uncertainties, masses, conventions, and provenance.¹¹

Although rotational constants may be fitted directly, the refinement uses principal moments of inertia as default observables, because they give a smoother geometry dependence and avoid reciprocal amplification:

$$I_{\alpha}^{(s)} = \frac{K}{B_{\alpha,e}^{(s)}}, \quad (19)$$

where K is the standard rotational conversion constant.

For planar molecules only two principal components are independent. If the molecular plane is the ab plane,

$$I_c = I_a + I_b, \quad (20)$$

Fitting all three components would duplicate one constraint. The workflow therefore uses one pair, AB , AC , or BC , and reports the omitted component as a diagnostic. If the pair is not specified, all admissible pairs are tested; the selected pair maximizes rank and minimizes the propagated instability of the omitted moment.

The selected observables and uncertainties for all isotopologues form the experimental part of the least-squares objective.

F. Least-Squares Refinement

As usual,^{1,7} the semiexperimental refinement is formulated as a weighted nonlinear least-squares problem. After hard constraints and parameter classes have been projected, the actual independent least-squares variables are collected in $\mathbf{p}$.

The residual vector is

$$\mathbf{r} = \mathbf{y}^{\text{obs}} - \mathbf{y}^{\text{calc}}(\mathbf{p}), \quad (21)$$

where $\mathbf{y}$ collects all experimental observables and predicate observations included in the refinement.

The weighted objective is

$$\chi^2 = \mathbf{r}^T \mathbf{W} \mathbf{r}, \quad (22)$$

where $\mathbf{W}$ contains the inverse variances of experimental and predicate observations. Robust fits reweight complete isotopologue blocks rather than individual rotational constants, while predicates retain their assigned statistical weights.

The optimization is performed using a weighted Levenberg–Marquardt procedure,

$$(\mathbf{J}^T \mathbf{W} \mathbf{J} + \lambda \mathbf{I}) \Delta \mathbf{p} = \mathbf{J}^T \mathbf{W} \mathbf{r}, \quad (23)$$

where $\mathbf{J}$ is the Jacobian matrix of the observables with respect to the independent variables $\mathbf{p}$ and λ is the damping parameter. Accepted steps are checked against the original topology, point group, irreducible-representation counts, and coordinate-family counts; topology-changing steps are rejected.

Convergence requires both parameter corrections and objective reduction to fall below thresholds. Diagnostic leave-one-isotopologue-out refits, small singular values, reduced rank, downweighted isotopologues, unstable planar pairs, and large propagated coordinate uncertainties are reported without redefining the coordinate model.

Following convergence, the covariance matrix in the independent fitted variables is obtained from

$$\mathbf{C}_p = \sigma_r^2 (\mathbf{J}^T \mathbf{W} \mathbf{J})^{-1}, \quad (24)$$

where σ_r^2 is the reduced residual variance.

G. Error Propagation

Uncertainty propagation follows the geometry-update chain. If $\mathbf{N}$ is the null-space basis of the hard constraints and class reductions, the covariance in the active coordinate space is

$$\mathbf{C}_a = \mathbf{N} \mathbf{C}_p \mathbf{N}^T. \quad (25)$$

The corresponding Cartesian covariance is

$$\mathbf{C}_x = \mathbf{T}_A \mathbf{C}_a \mathbf{T}_A^T. \quad (26)$$

For a GIC refinement, $\mathbf{T}_A = \mathbf{B}_Q^+$; for a Cartesian refinement, $\mathbf{T}_A = \mathbf{C}_{A_1}$. Thus both coordinate choices use the same covariance propagation.

Reported structural quantities are functions of the optimized Cartesian geometry. For a vector of quantities $\mathbf{g}(\mathbf{x})$, which may contain bond lengths, valence angles, dihedral angles, out-of-plane coordinates, ring descriptors, or user-defined primitive functions, the covariance is

$$\mathbf{C}_g = \mathbf{G}_x \mathbf{C}_x \mathbf{G}_x^T, \quad \mathbf{G}_x = \left. \frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right|_{\mathbf{x}_{\text{opt}}}. \quad (27)$$

Standard errors and correlations are obtained from $\mathbf{C}_g$. Hard constraints have no variance along removed directions, whereas predicate uncertainties enter the weights and contribute to $\mathbf{C}_p$.

The report lists propagated structural errors, correlations, SVD diagnostics of the final weighted Jacobian, projected constraints and classes, and matched active-coordinate labels. Final bond lengths, angles, dihedrals, and ring parameters are always evaluated from the optimized Cartesian geometry, so the reported tables are independent of whether the fit used GIC or Cartesian amplitudes. A numerical Hessian in the final independent variable space gives the stationary-point diagnostic.

H. Kraitchman Diagnostic

For singly substituted isotopologues, the workflow computes substitution coordinates from the Kraitchman equations²⁰ and compares their absolute values with the fitted coordinates in the parent principal-axis frame. Kraitchman coordinates are widely used in spectroscopic structure work, either directly as r_s substitution coordinates, as checks on r_0 or semiexperimental structures, or as practical atom-assignment diagnostics. In this workflow they are intentionally kept outside the least-squares objective: they are useful checks, but they do not retain coordinate signs and do not exploit the full correlated model. At the same experimental-information level and computational cost, the SE+fit route is therefore more informative because all substituted constants enter one constrained Cartesian fit, uncertainties are propagated through the same model, and the omitted planar component can remain an external diagnostic.

III. IMPLEMENTATION AND REPRODUCIBILITY

The implementation is organized around two reusable services. The coordinate-definition service builds the molecular coordinate model from Cartesian atoms and nuclear charges, assigning point-group symmetry and irreducible representations when requested. A separate B-matrix service evaluates the same frozen coordinate definition for any compatible Cartesian geometry. During refinement, the engine reuses the stored coordinate model and updates only the derivative information needed for the current step.

The standard input contains the Cartesian parent geometry, optional primitive freeze or constraint records, and isotopologue data with rotational constants, corrections, uncertainties, masses, and conventions.

Each run records the final Cartesian structure, propagated primitive parameters, rotational residuals, covariance and SVD diagnostics, projected constraints/classes, provenance, warnings, and minimum-check Hessian eigenvalues. These outputs provide the audit trail for

the algorithmically generated model and are the numerical source for the benchmark tables.

IV. RELATIONSHIP TO EXISTING APPROACHES

The framework combines established semiexperimental least-squares machinery^{7,10,11} with an algorithmic model-generation layer. The statistical model uses moments of inertia, vibrational and electronic corrections, predicates, parameter classes, leverage and correlation diagnostics, robust weighting, and final minimum checks. The new element is the formal definition and audit trail of the refinement variables, constraints, and parameter-class structure. Existing approaches automate the solution of a model; this work formalizes the generation of the model itself, and then solves it with the same statistical discipline.

Z-matrices remain compact for tree-like molecules, as illustrated below by glycolaldehyde. Their limitations become visible for rings, fused systems, and bridged frameworks: a molecular graph with cycles cannot be represented by a single reference tree without making some chemically equivalent quantities primary and others derived. In a least-squares refinement this affects not only notation but also the covariance attached to reported parameters.

Generalized internal coordinates are a flexible alternative and are widely used in optimization and vibrational analysis.^{13,15–18,21} In those workflows the generalized coordinates are usually a user-defined layer. Here they are generated from topology-based primitive coordinates, reduced to a non-redundant space, assigned to irreducible representations, and accessed through primitive-coordinate constraints. Kisiel’s STRFIT/PROSPE route⁸ provides another important reference point for rotational-structure fitting, but the structural model is again supplied by the user and may leave derived ring quantities outside the primary fitted covariance matrix.⁹

The practical distinction is summarized in Table I. The workflow retains the usual semiexperimental inputs and diagnostics while moving coordinate construction, symmetry completion, constraint and class generation, and covariance propagation to a common topology-derived layer. The user still controls the chemical problem and the quality of the input data; what is removed is the need to hand-build the non-redundant refinement model.

A. Computational Reference Data

The electronic-structure quantities used in this work were generated with the hierarchical family of Pisa composite schemes (PCSs). In most cases, vibrational corrections and constraint targets were taken from the double-hybrid member of the family (DPCS3), which has been validated in previous work for accurate structures and

fast rovibrational corrections. The only exception is conformer I of glycine, where the best available reference was used: PCS4, based on explicitly correlated CCSD(T)-F12 calculations close to the complete-basis-set limit with an additive core–valence correction. Outside this role as reference information, the electronic-structure data do not define the fitted coordinates; they enter only through explicit constraints, predicates, parameter classes, and ΔB_{vib} corrections.

V. RESULTS AND DISCUSSION

The systems are discussed in the order in which they test increasingly demanding parts of the same algorithmic model-generation problem. Glycolaldehyde is the acyclic reference case; cyclopentadiene tests ring closure; nitrobenzene and azulene test planar-component selection and fused aromatic topology; norcamphor tests bridged graphs and hydrogen-frame constraints. The anhydride series then probes how published data sets with different isotopic completeness are converted into equivalent input models. Glycine I and II test reduced-dimensionality constraints. Testosterone is kept separate and last because it is a scale demonstration rather than an isotope-rich benchmark. In all cases the input is Cartesian and no molecule-specific coordinate code, manual reference tree, or manual active-space selection is used.

Figure 3 is the central validation of the framework: it quantifies how topology perception, symmetry analysis, isotopic coverage, and reference information are transformed algorithmically into the final refinement model. The figure is therefore a model-generation audit, not simply a residual summary: it reports how many primitive/non-redundant coordinates are produced by the algorithmic generator, how many remain active after symmetry selection, how many survive constraint projection, and how many constraints are introduced without manual parameter selection. The rotational residuals remain in the expected semiexperimental range for standard, cyclic, planar, fused, bridged, and glycine benchmarks, but the central result is that these residuals are obtained from generated, auditable models. The glycine examples also separate two roles of theory: Gly-I starts from a reference that already satisfies the practical rotational-accuracy target, whereas Gly-II uses QC-derived relations to define missing reduced-dimensionality information.

Figure 3 also makes explicit what is generated. For azulene, topology and symmetry generate 48 final GICs, identify 17 totally symmetric coordinates, and reduce the fit to 10 effective variables without a manual choice of refined coordinates. For norcamphor, the same machinery constructs a non-redundant 48-coordinate GIC model for a low-symmetry bridged system, then generates 30 hydrogen-frame constraints and leaves 18 effective variables. These are the cases in which manual model formu-

TABLE I. Compact operational comparison of representative structure-refinement routes. The emphasis is model formulation rather than the statistical quality of a particular fit.

Feature	Z-matrix	User GIC or PROSPE	MORPHEUS
Input model	Internal tree	User model	Cartesian topology
Active variables	Manual	Manual	Auto rank-pruned
Rings/bridges	Tree-dependent	User-defined	Auto primitives
Missing isotopes	Manual constraints	Manual classes	Advisor constraints
Symmetry	In model	User supplied	Detected and audited
Output audit	Supplied model	Workflow dependent	Manifest and covariance

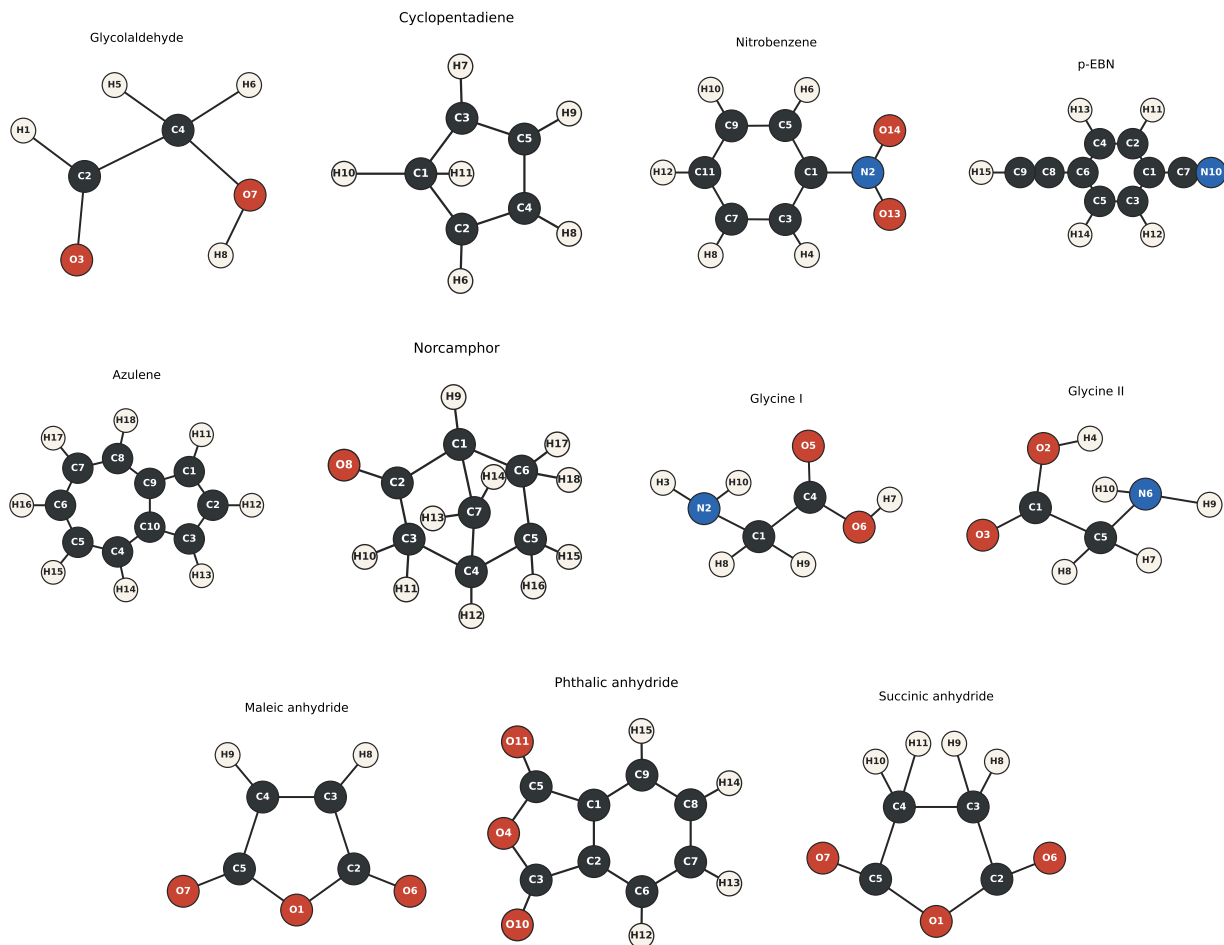


FIG. 2. Structures and atom numbering for the validation systems used to test algorithmic model generation: glycolaldehyde, cyclopentadiene, nitrobenzene, p-EBN, azulene, norcamphor, glycine I, glycine II, and the maleic, phthalic, and succinic anhydride series. For these systems, the same Cartesian numbering is used by topology perception, coordinate generation, constraints, diagnostics, and final structural tables.

lation is most exposed: fused rings and bridged frameworks make it difficult to decide by inspection which distances, angles, torsions, and hydrogen-frame quantities should be refined, tied, or constrained. Table II recasts the same information as model-formulation burden. The manual column lists the decision classes that would normally have to be settled before the least-squares fit; the generated column reports the resulting working model and residuals. Stretches, bends, torsions, out-of-plane terms, linear bends, and ring-deformation coordinates

are generated from the perceived graph and reduced in homogeneous families. Consequently, a stretching coordinate cannot replace a torsion or a ring coordinate during rank reduction.

The agreement with reference-quality refinements is therefore obtained after removing the molecule-specific model-design step, not after tuning it.

The three anhydrides were selected as a homologous literature-data series, not as unrelated examples. They have the same chemical motif but progressively different

Algorithmic coordinate generation and reduction

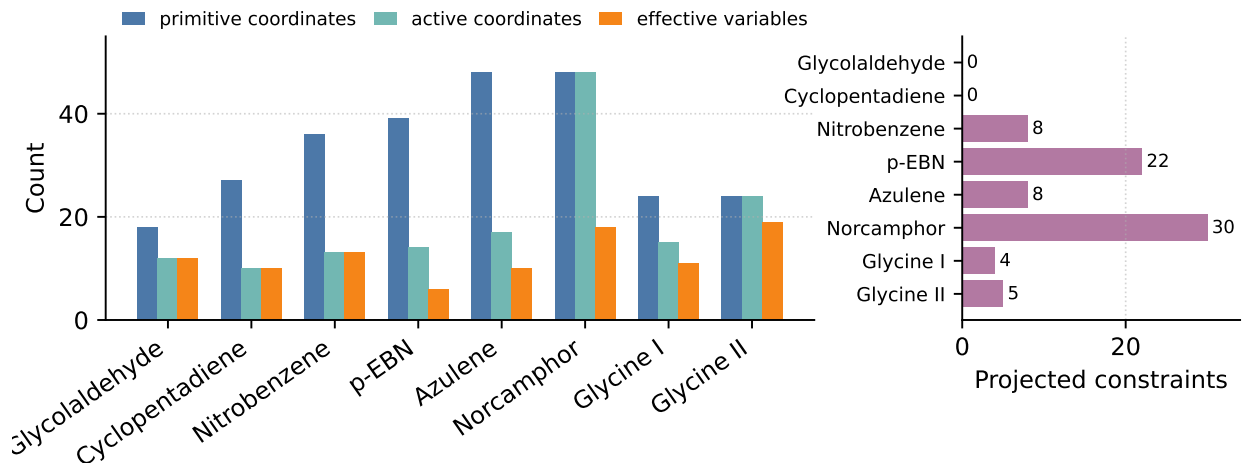


FIG. 3. Algorithmic model-generation audit for the representative refinements. The left panel reports the number of coordinates generated algorithmically, the symmetry-active subset before constraint projection, and the final least-squares variables after reduction. The right panel reports the number of constraints or frozen primitive parameters projected by the model layer. Manual coordinate-model edits are zero for every case. Detailed fit residuals and ranks are collected in the Supporting Information.

TABLE II. Decision burden removed by algorithmic model generation. Manual setup time is not reported because it depends on operator choices; the table therefore reports the reproducible decision burden. The final column counts molecule-specific coordinate or constraint edits made by the user in the reported runs.

Case	Manual workflow would decide	Generated model	Constraints/classes	Result	Manual edits
Azulene	fused-ring coordinates; symmetry; active variables; planar pair	48 GICs; 17 symmetric; 10 variables	8 primitive constraints	0.073 MHz RMS	0
Norcamphor	bridged coordinates; active variables; H-frame constraints; covariance targets	48 GICs; 48 active; 18 variables	30 H-frame constraints	0.071 MHz RMS	0
Glycine I	symmetry; unsupported H-frame directions; active variables	24 GICs; 15 A' ; 11 variables	4 QC constraints	0.0269 MHz RMS	0
Glycine II	low-symmetry active space; coupled descriptors; unsupported directions	24 GICs; 19 variables	5 QC constraints; coupled classes	0.249 MHz RMS	0

experimental completeness, and therefore test different parts of the workflow. Maleic anhydride is the fully determined case: the published work of Vogt, Demaison, and Rudolph²² reports the parent geometry, ground-state rotational constants, rovibrational corrections, and semiexperimental constants for a sufficiently rich isotopic set. Phthalic anhydride tests mixed estimation: Belyakov et al.²³ supplement the rotational data with weighted r_e^{BO} predicates. Succinic anhydride tests the opposite limit: Jahn et al.²⁴ provide only heavy-atom substitutions, so the hydrogen frame is not experimentally resolved and must be represented by explicit model relations.

This series also makes the input claim concrete. Once B_0 constants and rovibrational corrections ΔB_{vib} are available, the user transcribes the parent Cartesian geometry, the isotope substitutions, and the tabulated corrections. The program then perceives the topology and symmetry, constructs the non-redundant coordinate model, and the model layer applies any necessary constraints, predicates, or parameter classes as explicit statistical relations.

For maleic anhydride no auxiliary relation is introduced and 21 non-redundant symmetry coordinates are generated, of which the eight A_1 coordinates are the active variables for the planar SE fit (Table III). The GIC

and symmetry-Cartesian routes both give rank 8 and a rotational RMS residual of 0.036 MHz; the recovered bond lengths agree with the published r_e^{SE} values within 1.2×10^{-4} Å.

Table IV illustrates the type of chemically meaningful relations generated by the advisor. The user does not select active coordinates or construct a Z-matrix; instead, the program identifies unsupported structural directions and expresses them as primitive internal-coordinate relations, which are subsequently projected onto the algorithmically generated active space. The same representation also makes ring coordinates explicit: endocyclic bends and torsions are generated as linear combinations of primitive valence angles or dihedral angles rather than as molecule-specific coordinate choices.

For phthalic anhydride, the rotational information alone can give a small rotational residual while leaving the structural model rank deficient. The published solution is therefore naturally expressed as primitive-coordinate predicates with conservative uncertainties of 0.002 Å for bond lengths and 0.2° for angles. With these predicates, the fit has rank 14, condition number 4.05, and a rotational RMS residual of 0.50 MHz. It recovers the published r_e^{SE} structure within 0.0015 Å for the tabulated bond lengths and within 0.09° for the tabulated

TABLE III. Algorithmically generated symmetry-coordinate model for maleic anhydride from literature data. Only the totally symmetric A_1 block is active in the C_{2v} semiexperimental refinement.

Symmetry block	Coordinates	Role in the SE model
A_1	8	Active stretches/bends; rank 8
A_2	3	Torsion/out-of-plane diagnostics
B_1	3	Torsion/out-of-plane diagnostics
B_2	7	Stretch/bend diagnostics
Total	21	Non-redundant GIC space

TABLE IV. Chemically readable primitive-coordinate relations generated by the model layer. The first four entries are representative reduced-dimensionality constraints for glycine; the last two show the corresponding form of generated ring coordinates. Atom labels are used for readability.

Primitive relation	Generated relation	Role
O–H bond	$ROH = R(O,H)$	fixes unsupported O–H distance
C–O–H angle	$AOH = A(C,O,H)$	preserves hydroxyl orientation
H–N–H angle	$HNH = A(H,N,H')$	constrains amino frame
NH ₂ wag	$WNH_2 = OOP(H,N,H',C)$	removes unsupported wagging direction
Endocyclic bend	$RB = \sum c(i) A(i,j,k)$	defines ring valence coordinate
Endocyclic torsion	$RT = \sum c(i) D(i,j,k,l)$	defines ring puckering/torsional coordinate

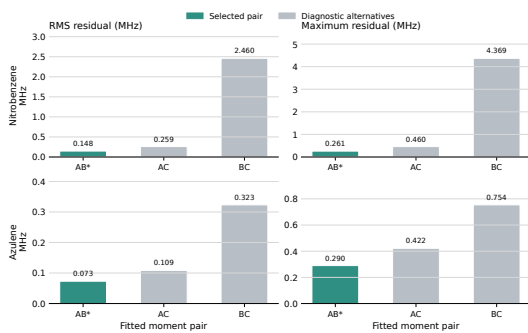


FIG. 4. Planar-component selection for nitrobenzene and azulene: the selected AB pair is highlighted and minimizes the complete post-fit diagnostic residual while the omitted component remains an external check. Detailed values are reported in the Supporting Information.

angles.

Succinic anhydride completes the series because only heavy atoms are isotopically substituted. The heavy-atom framework is determined, but the hydrogen positions are not independent spectroscopic observables. In this case a local hydrogen-frame constraint gives a stable fit with rank 5 and a rotational RMS residual of 0.035 MHz. Weighted C–H predicates are still useful diagnostically, but as soft observations they do not remove all unobserved hydrogen-frame directions; without an additional topological or constraint relation the optimizer can enter geometrically equivalent but chemically unacceptable valleys. The comparison makes the distinction operational: constraints remove unsupported directions, predicates add observations with declared uncertainty, and parameter classes tie equivalent corrections.

For planar molecules the model layer selects the best pair of rotational constants to be fitted, and retains the third rotational constant as an external check computed

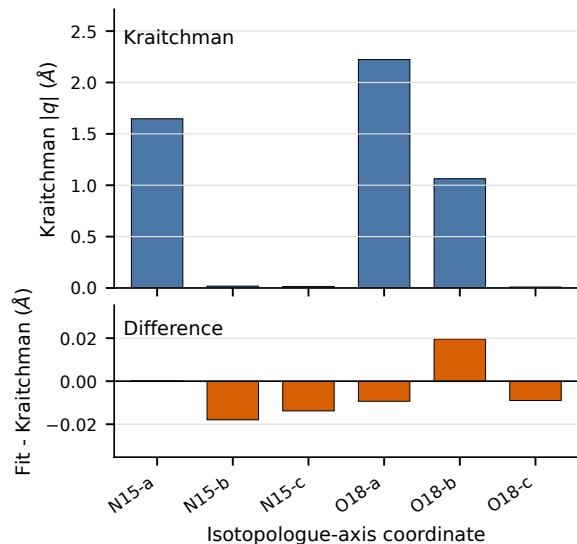


FIG. 5. Kraitchman diagnostic for the nitrobenzene planar benchmark. The upper panel reports absolute substitution coordinates from the Kraitchman analysis. The lower panel reports signed differences between the final SE coordinates and the Kraitchman values on an expanded scale. Numerical values are reported in the Supporting Information.

from the final Cartesian geometry. Figure 4 shows that the AB pair gives the smallest complete diagnostic residual for both nitrobenzene and azulene.

The next two diagnostics compare the fitted structures with independent structural information without changing the least-squares objective. The nitrobenzene Kraitchman comparison in Figure 5 confirms that the fitted Cartesian structure follows the substitution-coordinate pattern, while the fit itself remains based on moments of inertia. This is the intended use of Kraitch-

TABLE V. Effect of auxiliary model relations for phthalic anhydride. The predicate model uses the conservative r_e^{BO} uncertainties reported by Belyakov et al. and the same literature rotational data. Structural errors are maximum absolute deviations from the tabulated r_e^{SE} bond lengths or angles.

Auxiliary relation	Rank	Cond.	RMS	Error
None	13	∞	0.0065	3.13°
Frozen H frames	9	1056	0.0067	0.19°
r_e^{BO} predicates	14	4.05	0.50	0.0015 Å; 0.09°

man information here: it validates the SE+fit geometry but does not replace the statistically coupled refinement.

The Gly-I structural comparison in Figure 6 shows the scale of the semiexperimental correction when the starting QC geometry already reproduces the rotational data well. Since the reduced-dimensionality constraints are generated from the QC reference structure, the small SE–QC differences provide an a posteriori validation of the generated constraint model, not merely a check that the starting geometry was good.

Detailed atom numbering, rotational constants, topological parameters, Kraitchman diagnostics, reference-geometry comparisons, residuals, conditioning, and covariance diagnostics are reported in the Supporting Information. The main text retains the information needed to assess model formulation: rank, dimensionality, symmetry, constraint projection, planar-component choice, residual scale, and minimum checks.

A. Generated Reduced-Dimensionality Models

Incomplete isotopic coverage can leave some structural directions unsupported. The advisor therefore treats reduced dimensionality as part of the model-generation step. It preserves the highest compatible symmetry, identifies local coordinates that cannot be determined independently, and replaces them by primitive-coordinate constraints, predicates, or parameter classes.

Table VI shows that the reduced models are generated from isotope coverage and chemical equivalence rather than imported as a manual parameter subset. In glycine I, glycine II, and norcamphor the available substitutions do not support all hydrogen-related directions independently; the advisor therefore converts the unsupported directions into explicit primitive-coordinate relations derived from the reference structure.

B. Glycine as an End-to-End Model-Generation Test

The two glycine conformers test the same model-generation layer starting from QC geometries when hydrogen-related directions are only partly supported.

For glycine I, the parent, $^{13}\text{C}(\text{carboxyl})$, $^{13}\text{C}(\text{methylene})$, CD_2 , and ^{15}N isotopologues are fitted using the rotational constants and weights of Godfrey and Brown²⁵ and the zero-point corrections of Kasalová et al.²⁶ The QC reference already reproduces

the 15 vibrationally corrected constants with an RMS relative error of 0.0258% and a maximum error of 0.0363%. The advisor detects the C_s geometry and incomplete hydrogen information and generates four constraints: O–H distance, C–O–H angle, H–N–H angle, and NH_2 wag. Projection leaves 11 independent variables from 15 A' GICs, and the final fit gives a complete rotational RMS residual of 0.0269 MHz with a positive optimized-space Hessian.

The Gly-I comparison is intentionally stringent: the refinement starts from an already spectroscopically consistent QC structure rather than correcting a poor starting point. Since the reduced-dimensionality constraints are generated from the QC reference structure, the small SE–QC differences provide an a posteriori validation of the generated constraint model. The largest heavy-atom bond and valence-angle shifts relative to the final semiexperimental structure are 0.0034 Å and 0.25°, respectively, both within the scale expected for state-of-the-art QC structures and vibrational corrections.^{2,11}

Glycine II is the non-symmetric reduced-dimensionality case. The available isotopologues do not determine a stable full structure, so the QC reference is used to generate coupled primitive-coordinate constraints: C–H and N–H differences, two methylene angular combinations, and an average H–N–C–C torsion. These constraints are then projected onto the 24-dimensional C_1 GIC space.

Table IX shows that QC-derived targets produce the same rank and essentially the same residual scale as the published Kasalová-type constraints. The final QC-constrained refinement has 19 effective variables, a condition number of 1.05×10^4 , and a complete rotational RMS residual of 0.2491 MHz. The largest uncertainties remain in the amino-hydrogen frame, as expected from the substitution pattern.

C. Testosterone: Large-Molecule Scale Demonstration

The validation benchmarks above were selected because they represent available experimental gold standards while spanning acyclic, cyclic, fused, bridged, planar, homologous, and reduced-dimensionality cases. The model-generation layer is not limited to these sizes. As a scale check, the framework was also applied to testosterone, using the published experimental ground-state rotational constants and vibrational corrections for the observed T1-H2[−]-gp conformer.²⁷ This is a useful large

Glycine I fitted structural parameters and shifts from the PCS4 reference

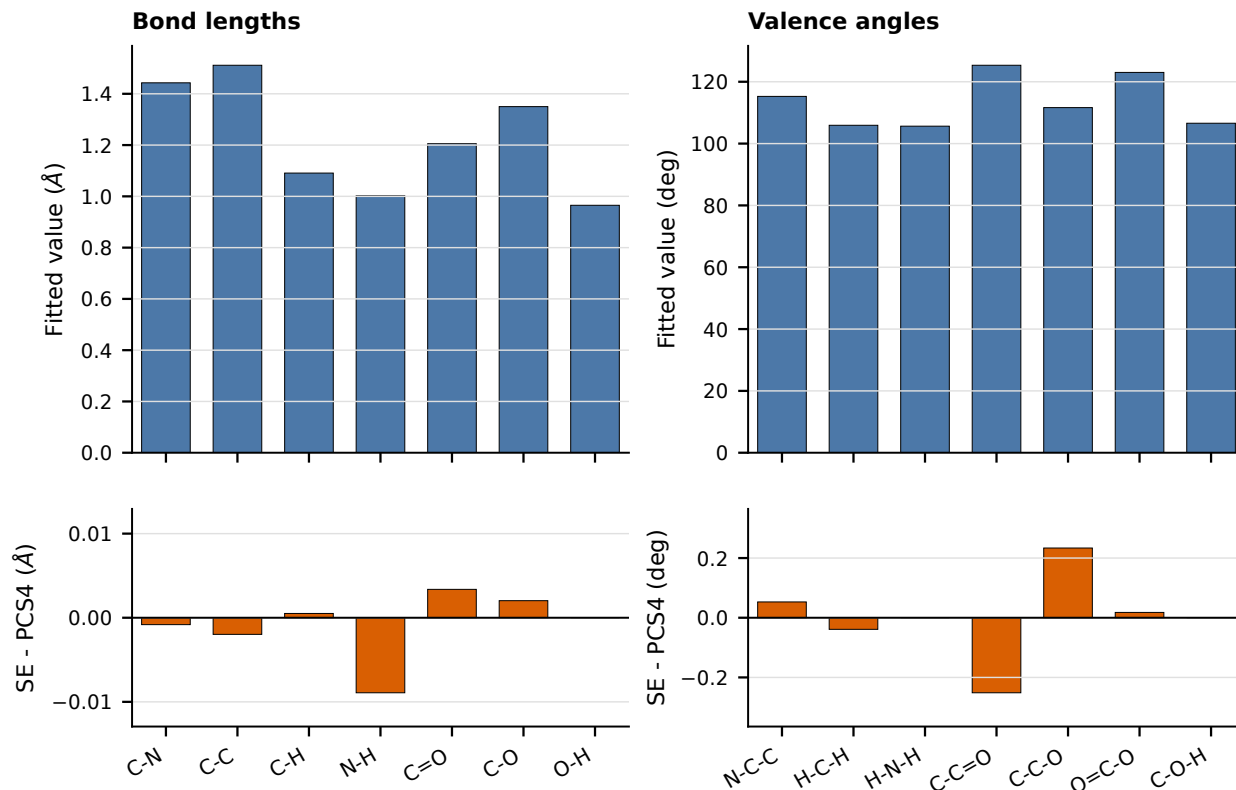


FIG. 6. Glycine I fitted structural parameters and shifts from the QC reference geometry. The upper panels report final SE bond lengths and valence angles; the lower panels report signed SE–QC differences on expanded scales.

TABLE VI. Autonomous structural-model decisions made by the advisor in the reduced-dimensionality benchmarks. “Active variables” are the independent least-squares coordinates after symmetry selection and projection of hard primitive-coordinate constraints.

System	GIC space	Variables	Auto constraints	Classes
Glycine I	24 GICs; 15 A'	11	4 QC functional constraints	none
Glycine II	24 GICs; C_1	19	5 QC functional constraints	coupled descriptors
Norcamphor	48 GICs; C_1	18	30 hydrogen-frame constraints	constrained H frames

TABLE VII. Glycine I rotational-constant diagnostics for the QC reference geometry and the final MORPHEUS semiexperimental geometry. Errors are computed against vibrationally corrected experimental constants for the five-isotopologue data set.

Geometry	RMS/MHz	Max/MHz	RMS/%	Max/%
QC reference	1.7414	3.4072	0.0258	0.0363
SE fit	0.0269	0.0571	0.0005	0.0011

molecule test because this level of treatment is rarely available for systems of this size: an accurate QC equilibrium geometry, accelerated ΔB_{vib} corrections, and algorithmic model generation are combined in one black-box protocol.

The unrefined QC+ ΔB_{vib} residuals with respect to

TABLE VIII. Parent glycine I rotational constants from QC and MORPHEUS/SE geometries. Corrected experimental and calculated values are in MHz.

Component	Exp. B_e	QC	QC–Exp.	SE	SE–Exp.
A	10420.1910	10422.9024	2.7114	10420.2481	0.0571
B	3907.4785	3907.5931	0.1146	3907.4786	0.0001
C	2934.6809	2935.7175	1.0366	2934.6808	-0.0001

the corrected experimental constants are 4.638, 0.674, and 0.563 MHz for A , B , and C , respectively. After a 141-dimensional non-redundant space is built, the class layer applies a single topology-derived C–C skeletal stretch class. Operationally, non-skeletal stretch combinations and the angular, torsional, and out-of-plane families are held fixed for this parent-only diagnostic,

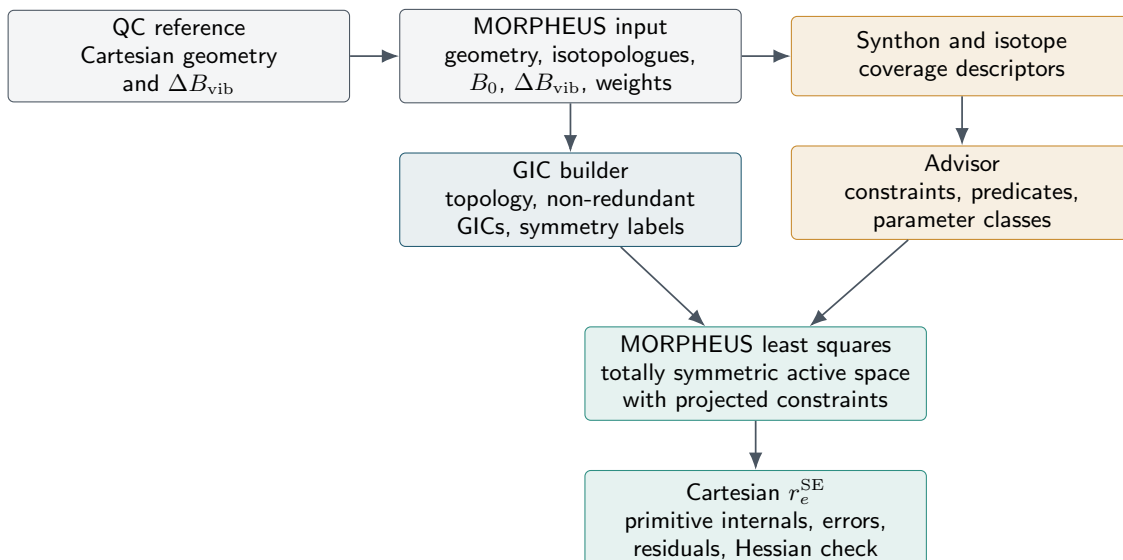


FIG. 7. End-to-end algorithmic glycine refinement starting from QC Cartesian reference geometries and vibrational corrections. The workflow uses the common job/xyzin container, chemical descriptors, and isotope coverage to generate the active coordinate and constraint model without manually selected fit variables.

TABLE IX. Glycine II constraint-source comparison. Both calculations used GIC coordinates, moments of inertia as observables, and the same 10 corrected Gly-II_n rotational data sets. Rotational residuals are reported over the diagnostic A , B , and C constants in MHz. The last column is the mean-square convention used for comparison with the published glycine refinement, not a second RMS value.

Constraint target source Rank	RMS	Max	$10^3 \langle \Delta B^2 \rangle$
Kasalová-type values	19 0.2344	0.8100	54.94
QC-derived values	19 0.2491	0.6707	62.06

while the retained C–C stretch block is replaced by one shared variable after rank reduction. The resulting refinement improves the residuals to -0.027 , 0.633 , and 0.291 MHz, reducing the RMS rotational residual from 2.73 to 0.40 MHz. This test is deliberately not presented as an isotope-rich structural validation: it demonstrates that a large-molecule QC geometry, vibrational corrections, non-redundant coordinate generation, and class-constrained refinement can be chained for a 49-atom fused-ring molecule, where sufficiently rich isotopic substitution data are not presently available for a full validation of the fit engine.

TABLE X. Testosterone T1-H2[−]-gp scale check. Residuals are relative to the corrected experimental equilibrium constants derived from the published B_0 and ΔB_{vib} values.

Model	Variables	ΔA	ΔB	ΔC
QC+ ΔB_{vib}	0	4.638	0.674	0.563
Class-constrained fit	1	-0.027	0.633	0.291

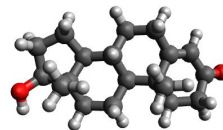


FIG. 8. Testosterone T1-H2[−]-gp large-molecule scale demonstration.

VI. CONCLUSIONS

The central contribution of this work is to move semiexperimental model formulation into a single auditable algorithm. Existing refinement methods generally assume that the coordinate model has already been supplied; here topology perception, symmetry assignment, homogeneous rank pruning, constraint projection, least-squares solution, and covariance propagation are part of one calculation rather than separate user decisions.

The benchmarks show that the resulting models remain topology-derived, symmetry-consistent, rank-consistent, and non-redundant for acyclic, cyclic, planar, fused, bridged, homologous anhydride, glycine, and large fused-ring cases. The program selects the active symmetry space, removes redundant coordinates, chooses the independent moment components for planar molecules,

generates constraints when isotopic coverage is incomplete, and propagates covariance to the reported structural parameters. Where both active coordinate models are applicable, GIC and symmetry-Cartesian refinements give the same effective dimensions and rotational residuals.

Electronic-structure information is used only where the experiment does not fully determine the structure. For Gly-I the QC reference already lies within the practical rotational-accuracy window, so the fit provides a statistically consistent semiexperimental correction and uncertainty estimate. For Gly-II and norcamphor, the advisor converts reference information and isotope coverage into explicit reduced-dimensionality constraints. The reference therefore supplies constraints for unsupported directions, while the spectroscopic data determine all supported directions.

This is not simply a convenient input generator. The final outputs remain ordinary chemical objects: Cartesian structures, bond lengths, valence angles, dihedrals, ring descriptors, propagated uncertainties, residuals, and diagnostics. What changes is that the path from topology and data to those objects is explicit, reproducible, and rank-audited, even for molecular graphs in which manual coordinate formulation is often the limiting step.

The testosterone scale check extends this logic to the current upper end of practical rotational-spectroscopic modeling. It does not claim a complete semiexperimental structure from a parent-only data set; instead, it shows that an accurate large-molecule QC geometry, accelerated ΔB_{vib} corrections, algorithmic non-redundant GIC generation, and class-constrained refinement can be chained for a 49-atom fused-ring molecule in a reproducible black-box calculation. This closes the loop from accurate large-molecule QC structures to algorithmically generated spectroscopic refinement models. The essential shift is from expert-dependent model formulation to reproducible algorithmic model generation.

SUPPORTING INFORMATION AVAILABLE

Atom-numbered structures for the benchmark and companion systems; complete rotational-constant comparisons for glycine I, glycine II, cyclopentadiene, nitrobenzene, and azulene; Kraitchman, reference-fit, and planar-component diagnostics; additional diagnostic tables for glycolaldehyde and norcamphor; final Cartesian coordinates, propagated topological parameters, residuals, and archived MORPHEUS output files.

¹J. Demaison, J. E. Boggs, and A. G. Császár, eds., *Equilibrium Molecular Structures: From Spectroscopy to Quantum Chemistry* (CRC Press, Boca Raton, 2011).

- ²C. Puzzarini and J. F. Stanton, *Physical Chemistry Chemical Physics* **25**, 1421 (2023).
- ³C. C. Costain, *The Journal of Chemical Physics* **29**, 864 (1958).
- ⁴F. Pawłowski, P. Jørgensen, J. Olsen, F. Hegelund, T. Helgaker, J. Gauss, K. L. Bak, and J. F. Stanton, *The Journal of Chemical Physics* **116**, 6482 (2002).
- ⁵M. Piccardo, E. Penocchio, C. Puzzarini, M. Biczysko, and V. Barone, *The Journal of Physical Chemistry A* **119**, 2058 (2015).
- ⁶E. Penocchio, M. Piccardo, and V. Barone, *Journal of Chemical Theory and Computation* **11**, 4689 (2015).
- ⁷M. Mendolicchio, E. Penocchio, D. Licari, N. Tasinato, and V. Barone, *Journal of Chemical Theory and Computation* **13**, 3060 (2017).
- ⁸Z. Kisiel, "PROSPE: Programs for ROTational SPEctroscopy," <https://info.ifpan.edu.pl/~kisiel/prospe.htm>, accessed 21 June 2026.
- ⁹A. Kanzow, B. Hartwig, P. Buschmann, K. G. Lengsfeld, C. M. Höhne, J. S. Hoke, F. Knüppe, F. Köster, J. K. van Spronsen, J. U. Grabow, D. McNaughton, and D. A. Obenchain, *Physical Chemistry Chemical Physics* **27**, 9491 (2025).
- ¹⁰L. S. Bartell, D. J. Romenesko, T. C. Wong, G. Sims, and L. E. Sutton, *Molecular Structure by Diffraction Methods* **3**, 72 (1975), *chemical Society Specialist Periodical Reports*.
- ¹¹M. Mendolicchio, L. Uribe, F. Lazzari, L. Crisci, G. Scalmani, M. J. Frisch, and V. Barone, *Physical Chemistry Chemical Physics* **27**, 16383 (2025).
- ¹²F. Lazzari, L. Crisci, and V. Barone, *Journal of Chemical Theory and Computation* **21**, 10617 (2025).
- ¹³E. B. Wilson, J. C. Decius, and P. C. Cross, *Molecular Vibrations: The Theory of Infrared and Raman Vibrational Spectra* (McGraw-Hill, New York, 1955).
- ¹⁴D. Cremer and J. A. Pople, *Journal of the American Chemical Society* **97**, 1354 (1975).
- ¹⁵P. Pulay and G. Fogarasi, *The Journal of Chemical Physics* **96**, 2856 (1992).
- ¹⁶C. Peng, P. Y. Ayala, H. B. Schlegel, and M. J. Frisch, *Journal of Computational Chemistry* **17**, 49 (1996).
- ¹⁷J. Baker, A. Kessi, and B. Delley, *The Journal of Chemical Physics* **105**, 192 (1996).
- ¹⁸V. Bakken and T. Helgaker, *The Journal of Chemical Physics* **117**, 9160 (2002).
- ¹⁹G. H. Golub and C. Reinsch, *Numerische Mathematik* **14**, 403 (1970).
- ²⁰J. Kraitchman, *American Journal of Physics* **21**, 17 (1953).
- ²¹A. V. Marenich, E. N. Brothers, H. P. Hratchian, and M. J. Frisch, *Journal of Chemical Theory and Computation* **21**, 10930 (2025).
- ²²N. Vogt, J. Demaison, and H. D. Rudolph, *Structural Chemistry* **22**, 337 (2011).
- ²³A. V. Belyakov, N. Vogt, J. Demaison, R. Y. Kulishenko, and A. A. Oskorbin, *Chemical Physics Letters* **795**, 139540 (2022).
- ²⁴M. K. Jahn, D. A. Obenchain, K. P. R. Nair, J.-U. Grabow, N. Vogt, J. Demaison, P. D. Godfrey, and D. McNaughton, *Physical Chemistry Chemical Physics* **22**, 5170 (2020).
- ²⁵P. D. Godfrey and R. D. Brown, *Journal of the American Chemical Society* **117**, 2019 (1995).
- ²⁶V. Kasalová, W. D. Allen, H. F. Schaefer, E. Czinki, and A. G. Császár, *Journal of Computational Chemistry* **28**, 1373 (2007).
- ²⁷L. Uribe, S. Di Grande, L. Crisci, F. Lazzari, M. Mendolicchio, and V. Barone, *The Journal of Physical Chemistry A* **128**, 2629 (2024).